CS162
Operating Systems and Systems Programming
Lecture 18

I/O Continued & Storage Devices

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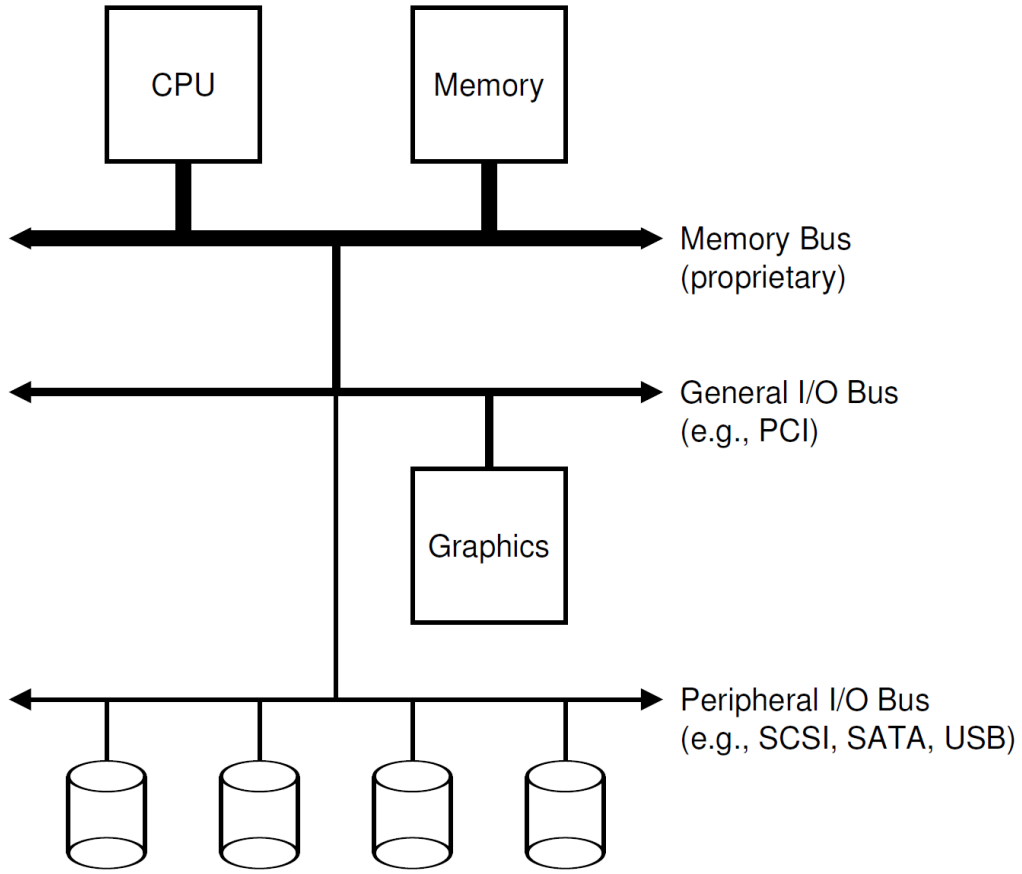
https://cs162.org/

Slides based on prior slide decks from David Culler, Natacha Crooks, John Kubiatowicz, Alison Norman and Lorenzo Alvisi

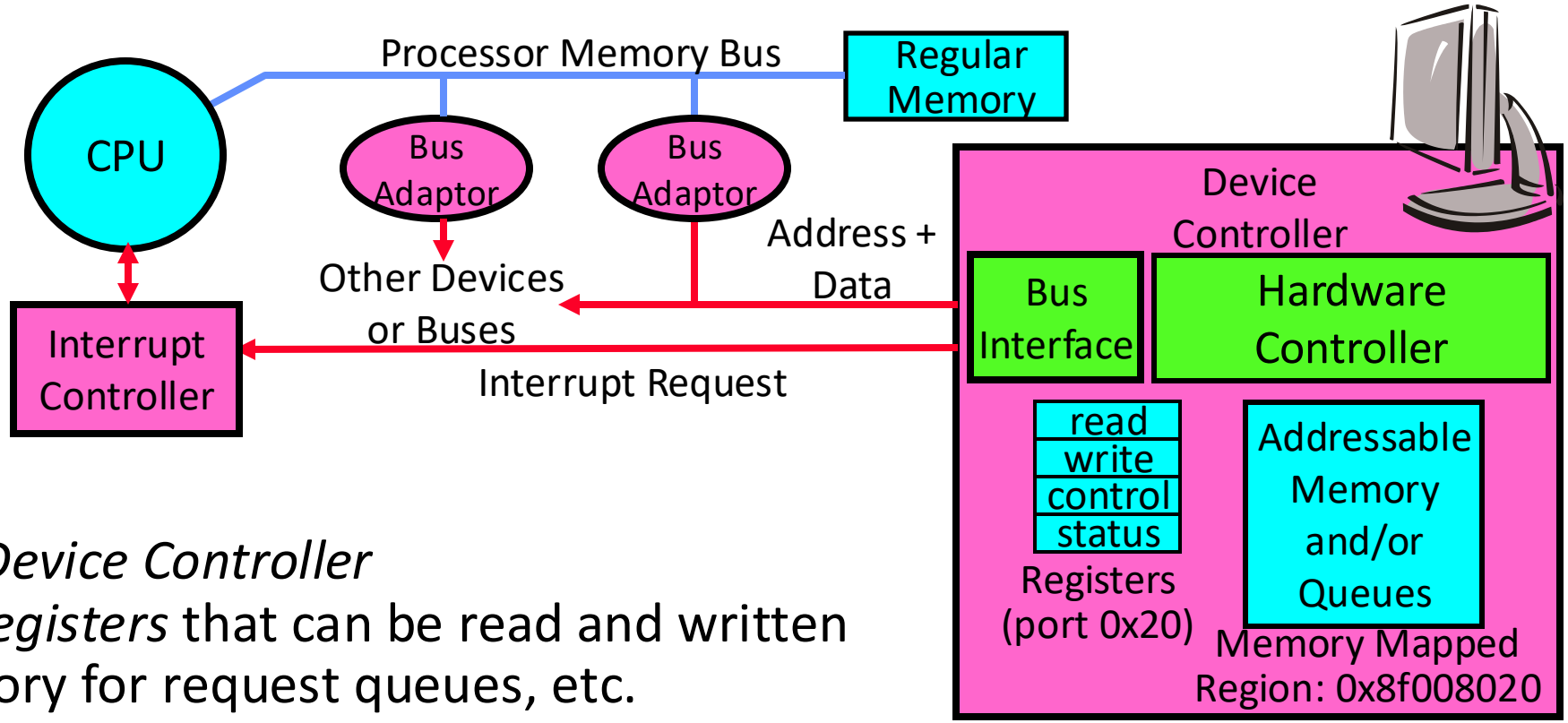
Recall : Simplified IO architecture



Follows a hierarchical structure because of cost:

the faster the bus, the more expensive

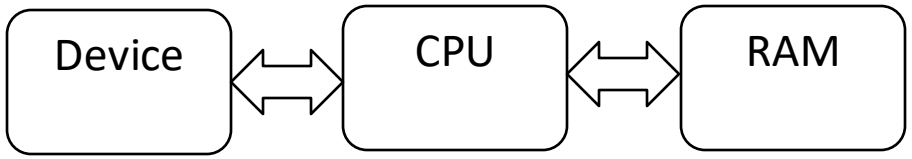
Recall: How Does Processor Talk to Devices?



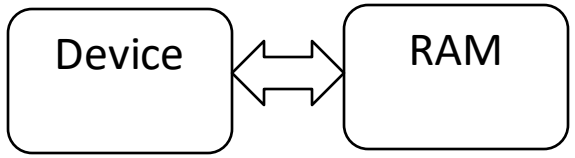
- CPU interacts with a *Device Controller*
  - Contains a set of *registers* that can be read and written
  - May contain memory for request queues, etc.
- Processor accesses registers in two ways:
  - Port-Mapped I/O: in/out instructions
    - » Example in Intel assembly: out 0x21,AL
  - Memory-mapped I/O: load/store instructions
    - » Registers/memory appear in physical address space

Recall: programmed I/O vs. direct memory access

- Programmed I/O:
  - CPU issues read request
  - Device interrupts CPU with data
  - CPU writes data to memory
  - Pros: simple hardware. Cons: Poor CPU is always busy!



- Direct-memory-access (DMA):
  - CPU sets up DMA request
    - » Gives controller access to memory bus
  - Device puts data on bus & RAM accepts it
  - Device interrupts CPU when done



How can the OS handle ~~one~~ all devices

- How do we fit devices with specific interfaces into the OS, which should remain general?
  - Build a “device-neutral” OS and hide details of devices from most of OS

- Abstraction to the rescue!
  - **Device Drivers** encapsulate all specifics of device interaction
  - Implement device-neutral interfaces

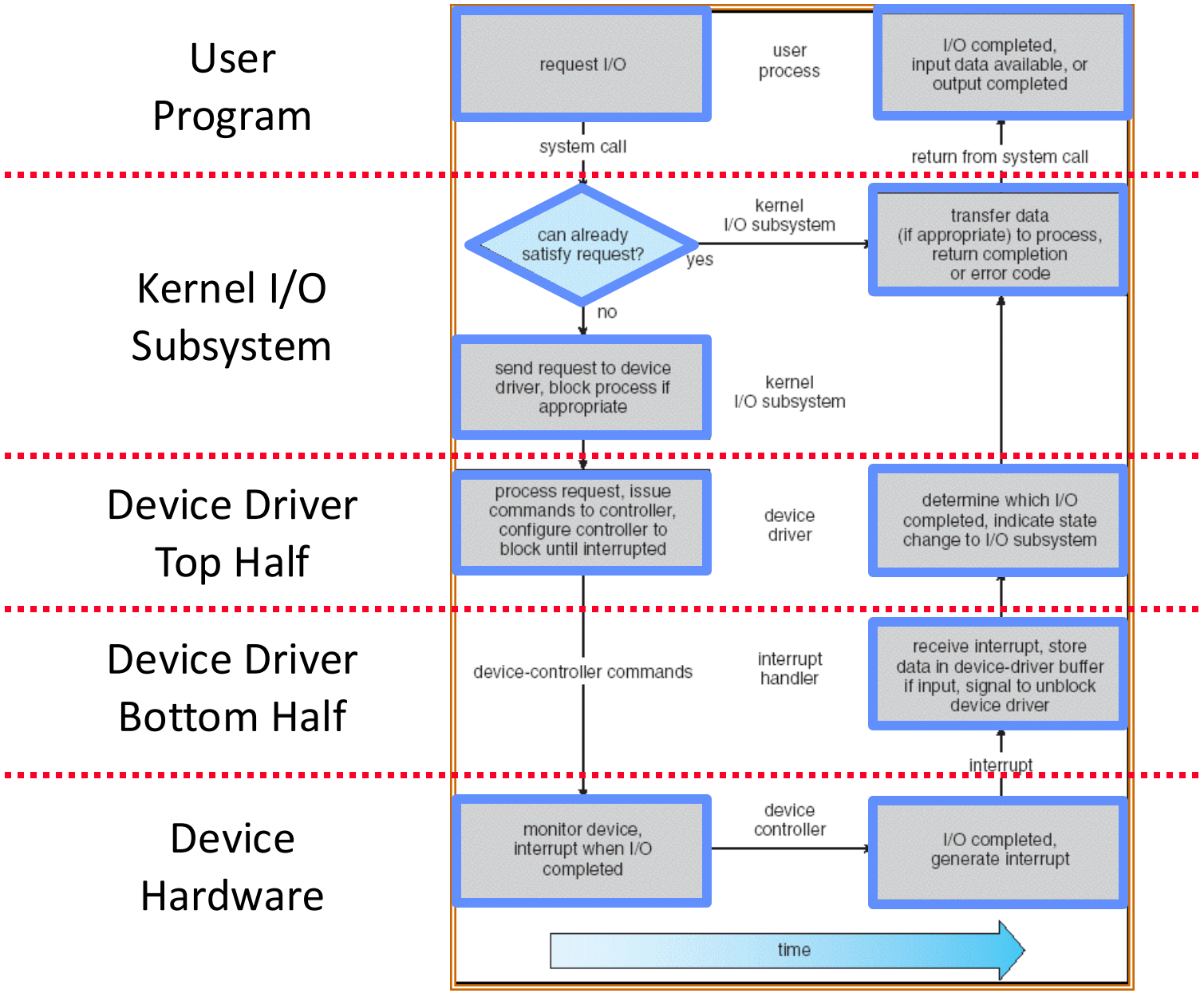
Device Drivers

- Device Driver: Device-specific code in the kernel that interacts directly with that device hardware
  - Supports a standard, internal interface
  - Special device-specific configuration supported with the `ioctl()` system call

- Device Drivers are typically divided into two pieces:
  - **Top half:** accessed in call path from system calls
    - » implements a set of standard, cross-device calls like `open()`, `close()`, `read()`, `write()`, `ioctl()`, `strategy()`
    - » This is the kernel's interface to the device driver
    - » Top half will *start* I/O to device, may put thread to sleep until finished
  - **Bottom half:** run as interrupt routine
    - » Gets input or transfers next block of output
    - » May wake sleeping threads if I/O now complete

- Your body is 90% water, your OS is 70% device-drivers

Putting it together: Life Cycle of An I/O Request



Summary

- I/O Devices Types:
  - Many different speeds (0.1 bytes/sec to 50+ GBytes/sec)
  - Different Access Patterns:
    - » Block Devices, Character Devices, Network Devices
- Device Controllers: Hardware that controls actual device
  - Processor Accesses through I/O instructions (port-mapped I/O) or load/store to special physical memory addresses (memory-mapped I/O)
- Notification mechanisms
  - Interrupts
  - Polling: Report results through status register that processor looks at periodically
- Device drivers interface to I/O devices
  - Provide clean Read/Write interface to OS above
  - Manipulate devices through PIO, DMA & interrupt handling

How Do User Applications Handle Timing?

- Blocking I/O interface: "Wait"
  - When request data (e.g. `read()` system call), put process to sleep until data is ready
  - When write data (e.g. `write()` system call), put process to sleep until device is ready for data
- Non-blocking I/O interface: "Don't Wait"
  - Returns quickly from read or write request with count of bytes successfully transferred
  - Read may return nothing, write may write nothing
- Asynchronous I/O interface: "Tell Me Later"
  - When request data, take pointer to user's buffer, return immediately; later kernel fills buffer and notifies user (or lets user check/poll)
  - When send data, take pointer to user's buffer, return immediately; later kernel takes data and notifies user (or lets user check/poll)

Storage and Filesystems

- Storage devices: how do we measure their performance and how do they work?
  - We'll focus on hard disks and solid-state drives (SSDs)

- File systems: most common abstraction on top of storage devices
  - Let users organize data into named files in hierarchical paths

Measuring Storage Device Performance

Latency: time to complete a task

- Measured in units of time (s, ms, us, …, hours, years)

Throughput or *Bandwidth*: rate at which tasks are performed

- Measured in units of things per unit time (bytes/s, ops/s)

Start up *or* *Overhead*: time to initiate an operation

Most I/O operations are roughly linear in *b* bytes

- Latency(b) = Overhead + b/TransferCapacity

Storage Devices

Magnetic disks (HDDs)

- Storage that rarely becomes corrupted
- Large capacity at low cost
- Block level random access (except for SMR – later!)
- Slow performance for random access
- Better performance for sequential access

Flash memory (SSDs)

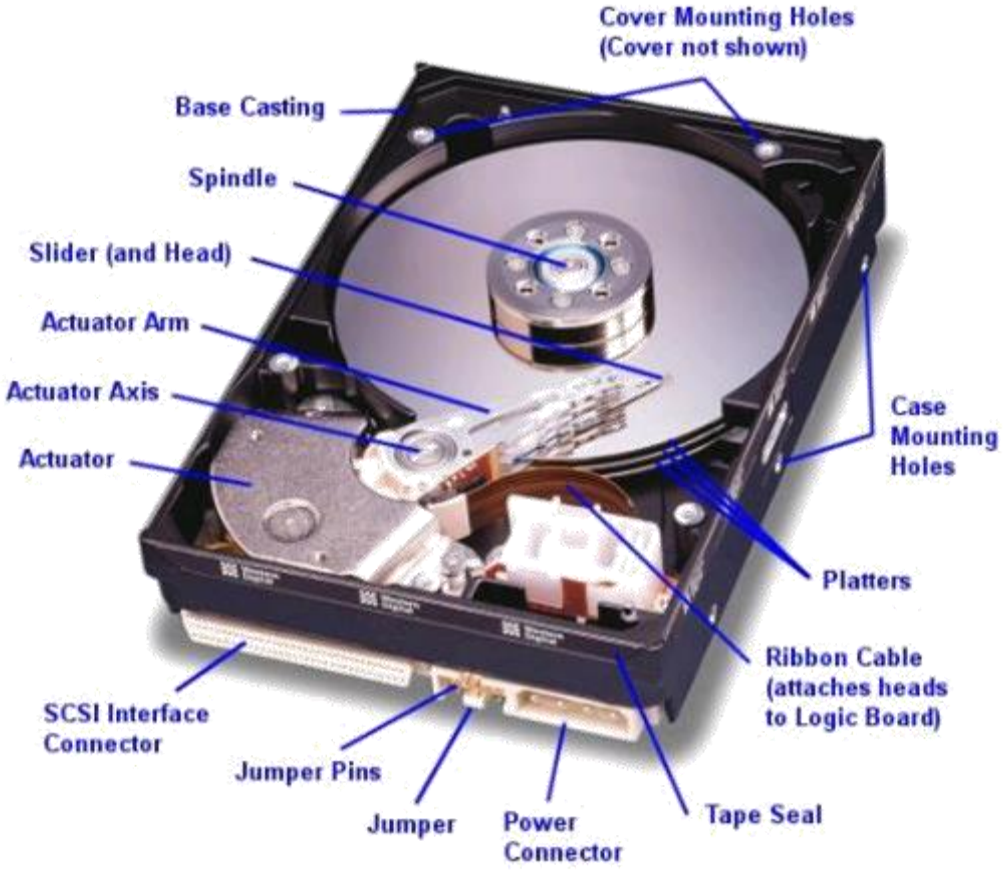
- Storage that rarely becomes corrupted
- Capacity at intermediate cost (2-20x disk)
- Block level random access
- Good performance for reads; worse for random writes
- Wear patterns issue

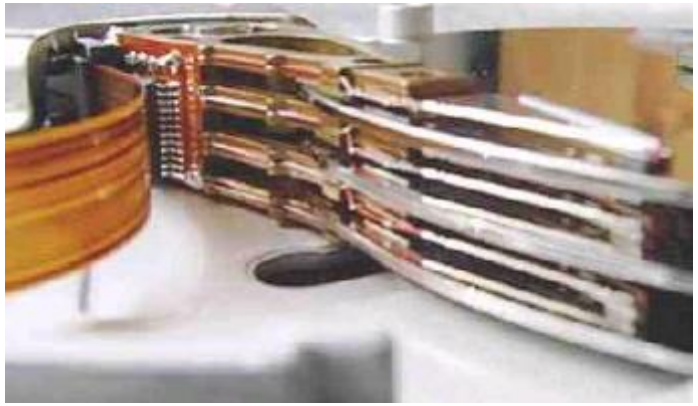
Hard Disk Drives (HDDs)

IBM Model 350, 1956

(first commercial HDD, 3.75 MB)

Hard Disk Drives (HDDs)

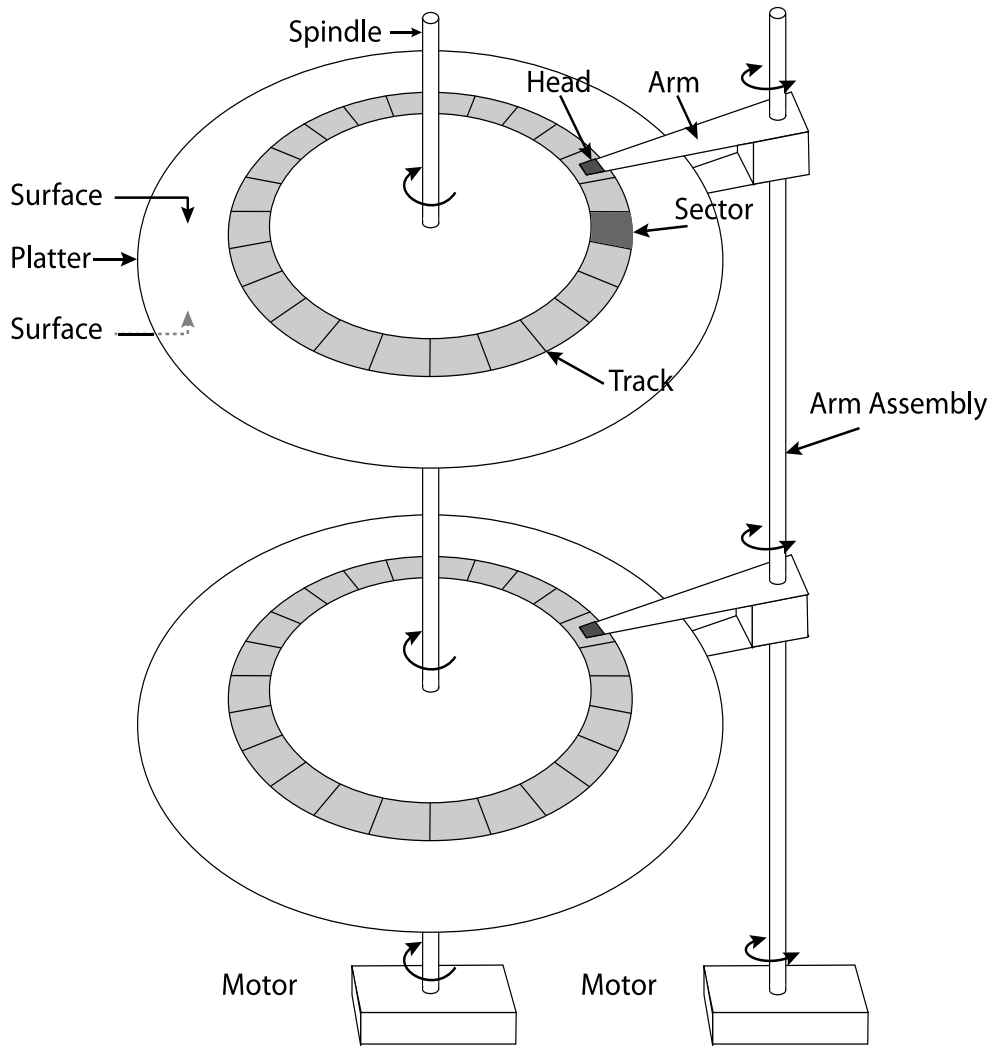




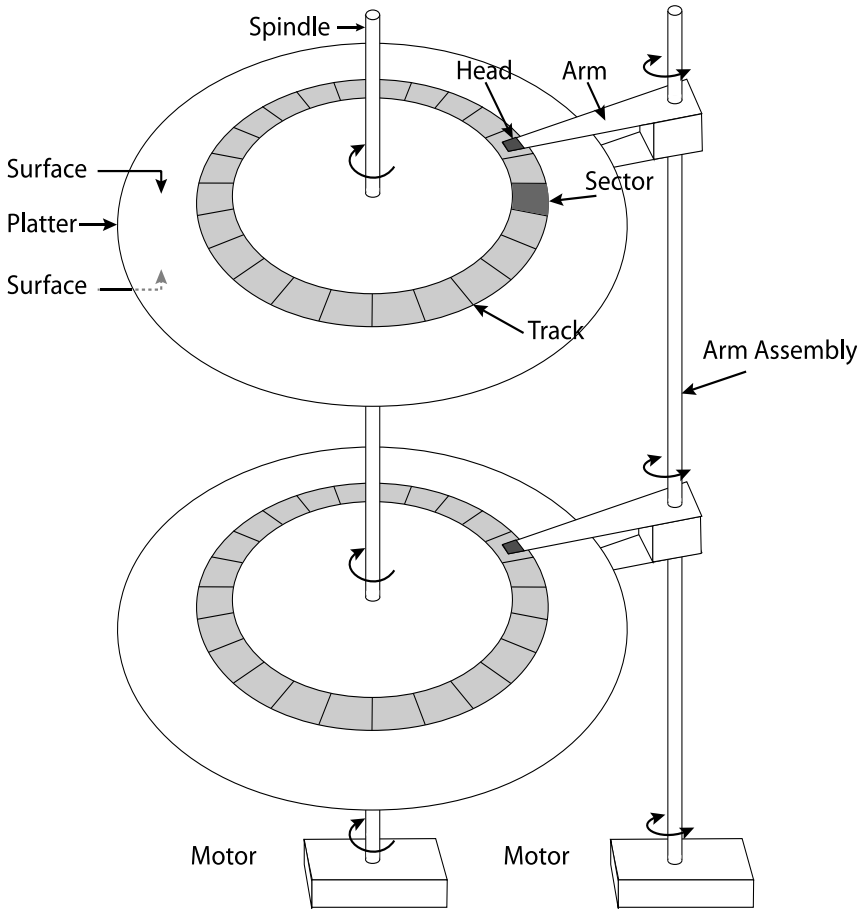
Read/Write Head Side View

The Amazing Magnetic Disk

Store data magnetically on thin metallic film bonded to rotating disk of glass, ceramic, or aluminum



The Amazing Magnetic Disk



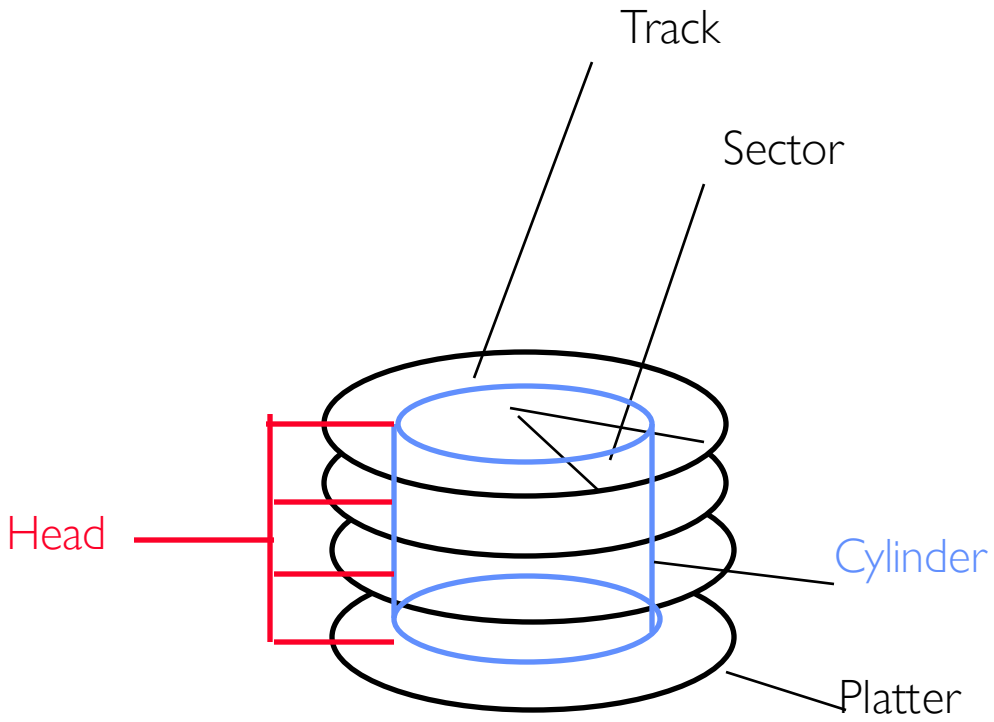
Track: concentric circle on surface

Sectors: slice of a track

- Smallest addressable unit
- Are units of transfers

Cylinder: all the tracks under the head at a given point, across all surfaces

The Amazing Magnetic Disk



Track lengths vary across disk: outside tracks have more sectors per track, higher bandwidth

Disk is organized into regions of tracks with the same number of sector/tracks

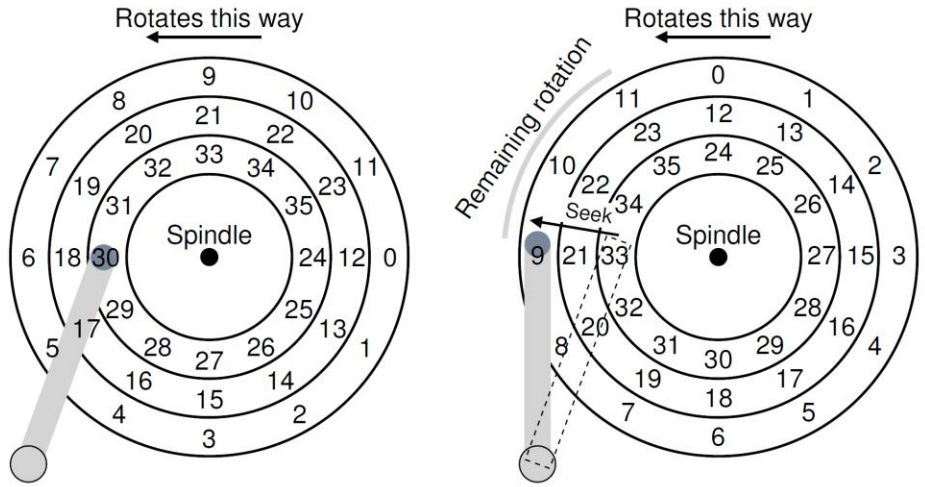
Usually, only outer half of radius is used

Reading/Writing Data

Seek time: position the head/arm over the proper track

Rotational latency: wait for desired sector to rotate under r/w head

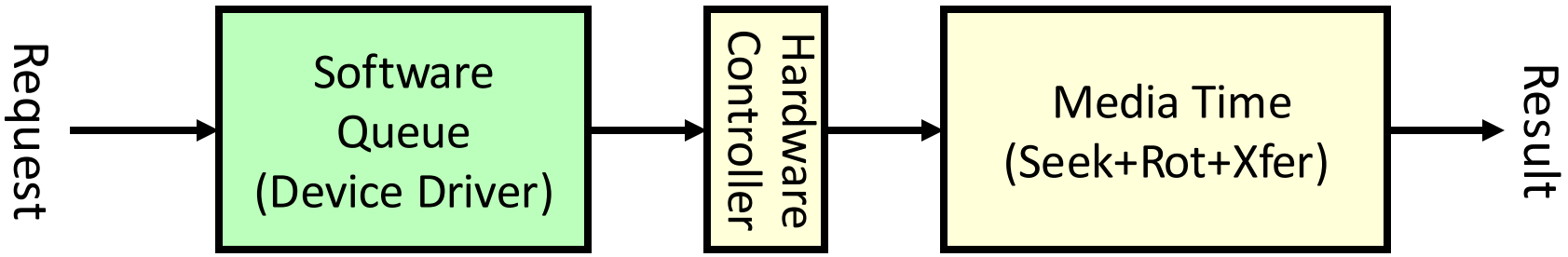
Transfer time: transfer a block of bits (sector) under r/w head



Reading/Writing Data

Request Time =

Queueing Time + Controller Time + Seek + Rotational + Transfer



Typical Numbers for Magnetic Disk

Parameter	Info/Range
Space/Density	Space: 18TB (Seagate), 9 platters, in 3½ inch form factor! **Areal Density: ≥ 1 Terabit/square inch! (PMR, Helium, …)**
Average Seek Time	Typically 4-6 milliseconds
Average Rotational Latency	Most laptop/desktop disks rotate at 3600-7200 RPM (16-8 ms/rotation). Server disks up to 15,000 RPM. Average latency is halfway around disk so 4-8 milliseconds
Controller Time	Depends on controller hardware
Transfer Time	Typically 50 to 250 MB/s. Depends on: • Transfer size (usually a sector): 512B – 1KB per sector • Rotation speed: 3600 RPM to 15000 RPM • Recording density: bits per inch on a track • Diameter: ranges from 1 in to 5.25 in
Cost	Used to drop by about 2x every 1.5 years, now slowing down

Disk Performance Example

Key to using disk effectively (especially for file systems) is to minimize seek & rotational delays

Do access patterns influence how fast can read/write to disk?

Assume a disk with avg seek time of 5ms, 7200RPM ⇒

Time for rotation: 60000 (ms/min)/7200(rev/min) ~= **8ms**

Assume transfer rate of 50MByte/s, block size of 4Kbyte ⇒

Time to read 1 sector: 4096 bytes/50×10^6 (bytes/s) = 81.92×10^{-6} sec ~= **0.082ms**

Disk Performance Example

Read block from random place on disk (random reads):

- Seek (5ms) + Rot. Delay (4ms) + Transfer (0.082ms) = 9.082ms
- Approx 9ms to fetch/put data: 4096 bytes/9.082×10^{-3} s $\cong$ 451KB/s

Read block from random place in same cylinder:

- Rot. Delay (4ms) + Transfer (0.082ms) = 4.082ms
- Approx 4ms to fetch/put data: 4096 bytes/4.082×10^{-3} s $\cong$ 1.03MB/s

Read next block on same track (sequential reads):

- Transfer (0.082ms): 4096 bytes/0.082×10^{-3} s $\cong$ 50MB/sec

When is Disk Performance Highest?

When there are big sequential reads, or

When there is so much work to do that they can be piggy backed (reordering queues—one moment)

OK to be inefficient when things are mostly idle

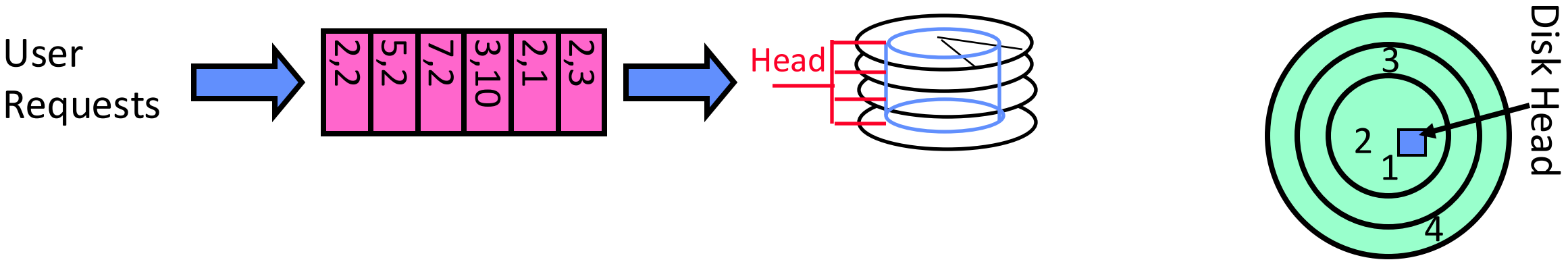
Bursts are both a threat and an opportunity

<your idea for optimization goes here>

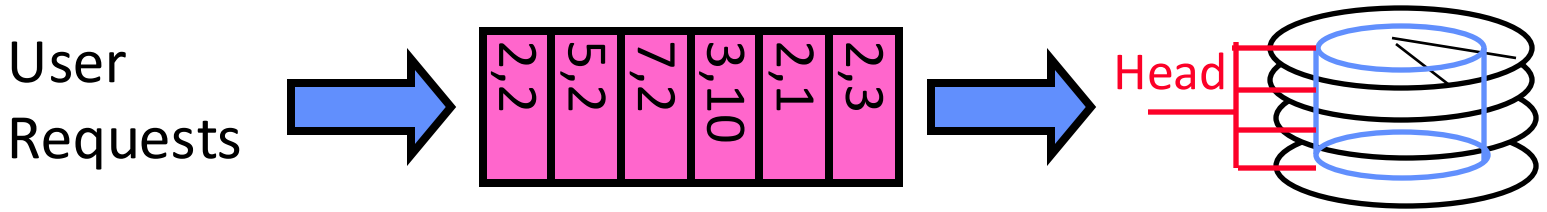
Waste space for speed?

Disk Scheduling (1/3)

Disk can do only one request at a time; What order do you choose to do queued requests?



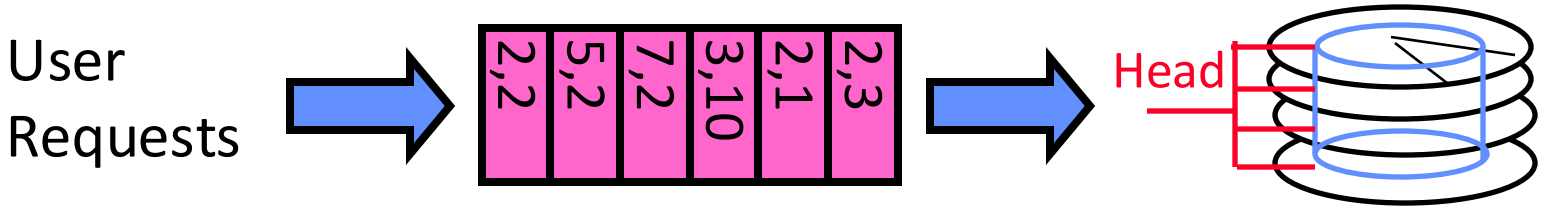
Disk Scheduling (1/3)



FIFO Order
Fair among requesters, but order of arrival may be to random spots on the disk

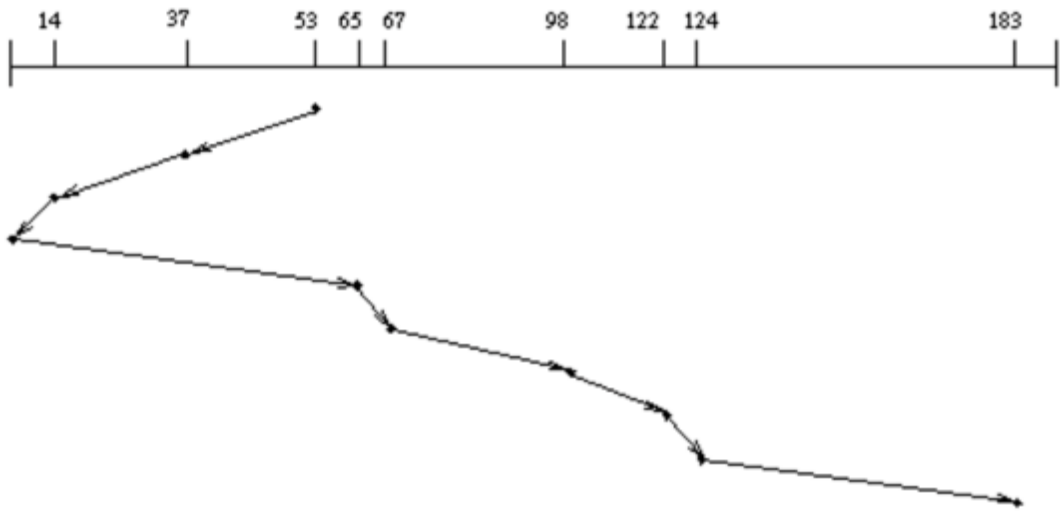
SSTF: Shortest seek time first
Pick the request that's closest on the disk
Con: SSTF good at reducing seeks, but may lead to starvation

Disk Scheduling (2/3)



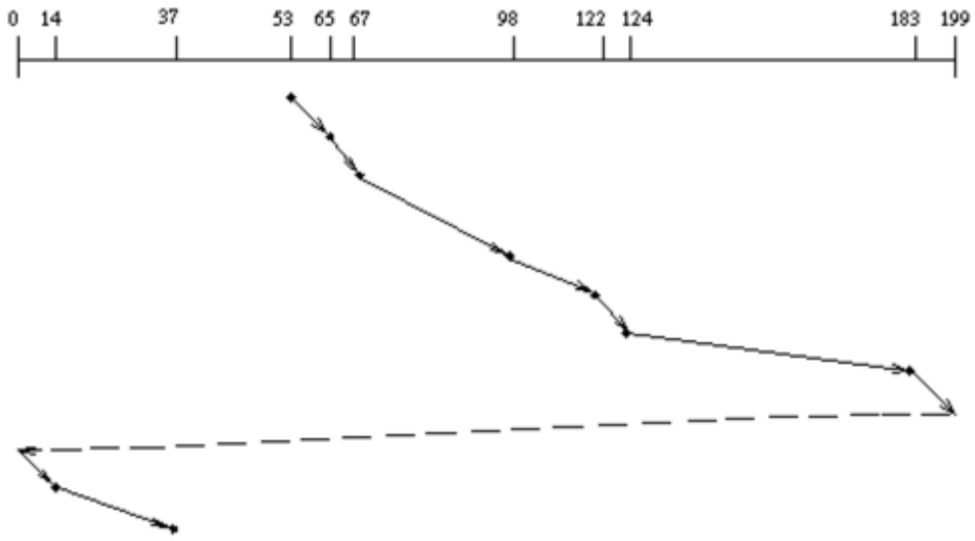
SCAN: Implements an Elevator Algorithm: take the closest request in the direction of travel

– No starvation, but retains flavor of SSTF



Disk Scheduling (3/3)

C-SCAN: Circular-Scan: only goes in one direction
– Skips any requests on the way back
– Fairer than SCAN, not biased towards pages in middle



Lots of Intelligence in the Device Controller

Shingled Magnetic Recording (SMR): pack more data with error correcting codes

Disk write head has wider field than read head, so overlap tracks

Hide corruptions due to neighboring track writes (but must fix in background)

Sector sparing

Remap bad sectors transparently to spare sectors on the same surface

Slip sparing

Remap all sectors (when there is a bad sector) to preserve sequential behavior

Track skewing

Sector numbers offset from one track to the next, to allow for disk head movement for sequential ops

Example of Current HDD

- Seagate Exos X24 (2023)
  - 24 TB hard disk
    - » 10 platters, 20 heads
    - » 1.26 TB/in^2
    - » Helium filled: reduce friction and power
  - 4.16 ms average seek time
  - 4096 byte physical sectors
  - 7200 RPMs
  - Dual 6 Gbps SATA /12Gbps SAS interface
    - » 285MB/s MAX transfer rate
    - » Cache size: 512MB
  - Price: $ 479 (~ $0.02/GB)
- Original IBM Personal Computer/AT (1986)
  - 30 MB hard disk
  - 30-40 ms average seek time
  - 0.7-1 MB/s (est.)
  - Price: $500 ($17K/GB)

800k x 🤩

385 x 🙂

10 x 😢

850k x 🤩

Solid State Drives (SSDs)

Invented by Fujio Masuoka at Toshiba in 1980

Modern flash memory in laptops, phones, etc:

- Sector (4 KB page) addressable, but stores 4-64 “pages” per memory “block”
- Trapped electrons distinguish between 1 and 0

No moving parts (no rotate/seek motors)

- Eliminates seek and rotational delay (0.1-0.2ms access time)
- Very low power and lightweight
- Limited “write cycles”

The Flash Cell

Encode bit by trapping electrons into a cell

Single-level cell (SLC)

Single bit is stored within a transistor

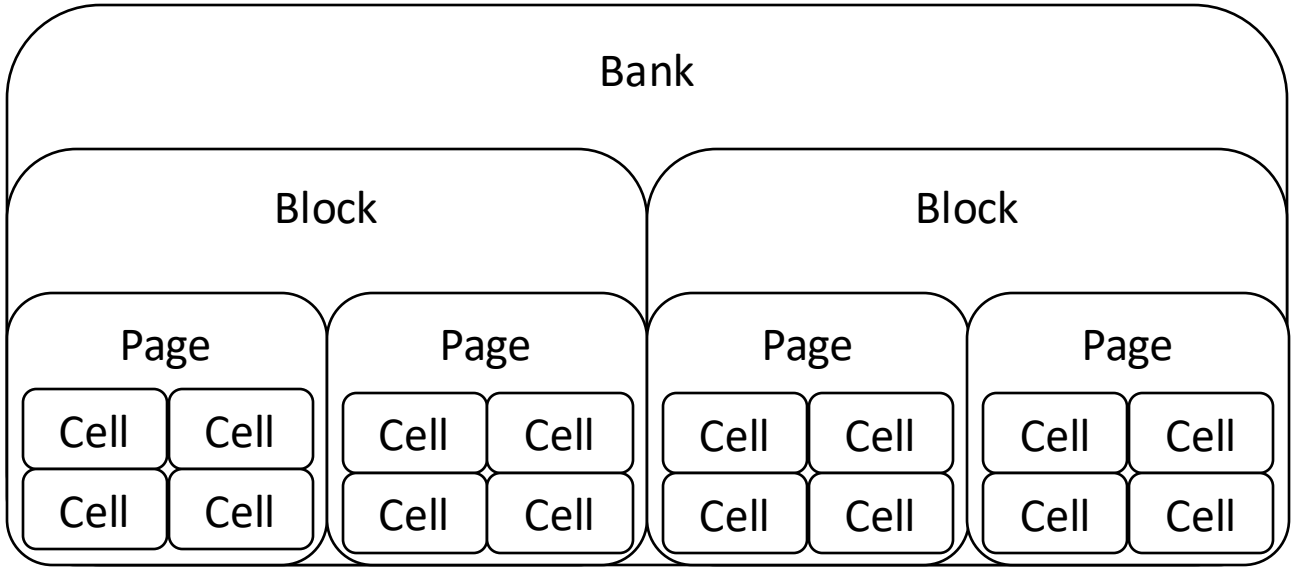
Faster, more lasting (50k to 100k writes before wear out)

Multi-level cell (MLC)

Two/three bits are encoded into different levels of charge

Wear out much faster (1k to 10k writes)

Of Banks, Blocks, and Cells



Flash chips organized in banks

- Banks can be accessed in parallel

Blocks 128 KB/256KB

- (64 to 258 pages)

⚠ Confusing: not same as OS and HDD "blocks"!

Pages Few KB

Cells 1 to 4 bits

Distinction between blocks and pages important in operations!

Low-level Flash Operations

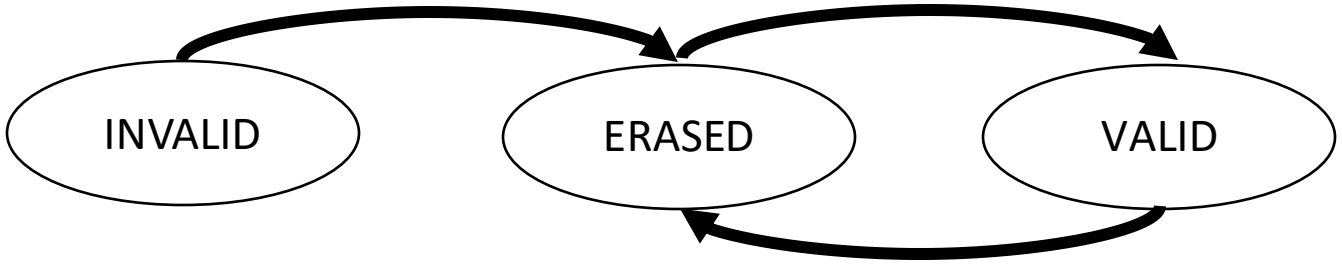
How do you read?

- Chip supports reading *pages*
- 10s of microseconds, independently of the previously read page

What about writing? More complicated!

- Must first *erase the block*
  - » Erase quite expensive (milliseconds)
- Once block has been erased, can then *program a page*
  - » Change 1s to 0s within a page.
  - » 100s of microseconds.
- Blocks can only be erased a limited number of times!

Low-level Flash Operations



		`iiii`	*Initial: pages in block are invalid* (`i`)
Erase()	→	`EEEE`	*State of pages in block set to erased* (`E`)
Program(0)	→	`VEEE`	*Program page 0; state set to valid* (`V`)
Program(0)	→	**error**	*Cannot re-program page after programming*
Program(1)	→	`VVEE`	*Program page 1*
Erase()	→	`EEEE`	*Contents erased; all pages programmable*

Low-level Flash Operations

Assume block of 4 pages. All valid.

Want to write Page 0

Page 0	Page 1	Page 2	Page 3
00011000	11001110	00000001	00111111
VALID	VALID	VALID	VALID

Step 1: erase full block

Page 0	Page 1	Page 2	Page 3
11111111	11111111	11111111	11111111
ERASED	ERASED	ERASED	ERASED

Step 2: program page 0

Page 0	Page 1	Page 2	Page 3
00000011	11111111	11111111	11111111
VALID	ERASED	ERASED	ERASED

SSD Architecture

Recall that SSDs uses low-level Flash operations to provide same interface as HDD

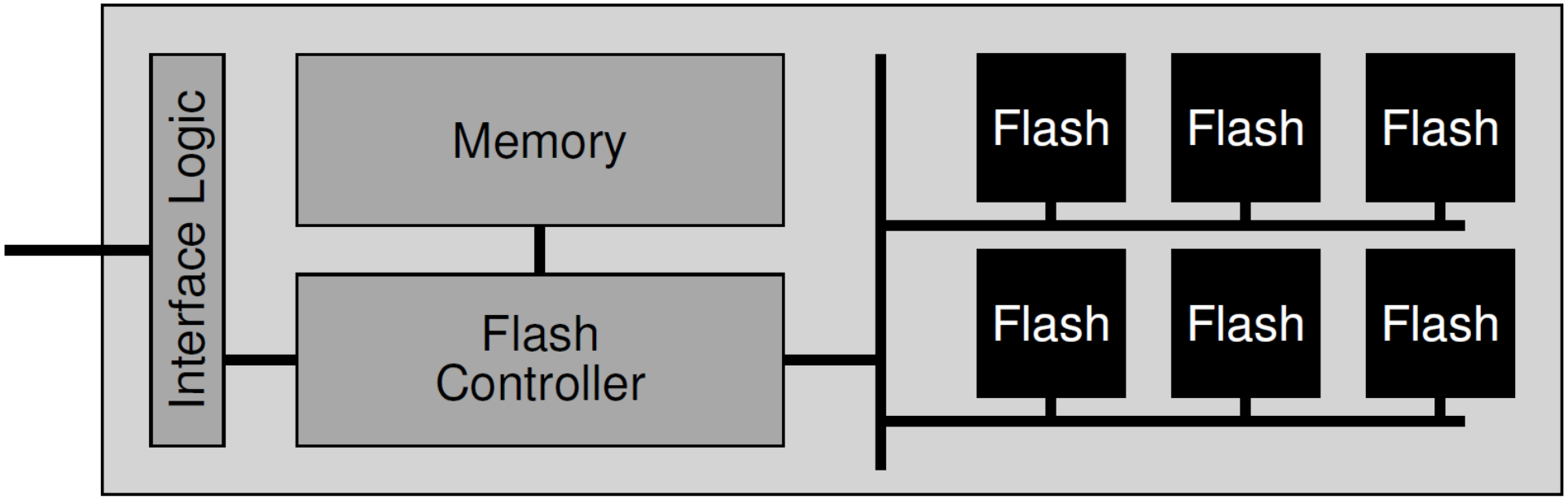
- read and write chunk (4KB) at a time

Reads are easy, but for writes, can only overwrite data one block (256KB) at a time!

Why not just erase and rewrite new version of entire 256KB block?

- Erasure is very slow (milliseconds)
- Each block has a finite lifetime, can only be erased and rewritten about 10K times
- Heavily used blocks likely to wear out quickly

SSD Architecture (Simplified)



Flash Translation Layer (FTL)

Add a layer of indirection: the flash translation layer

Translates request for logical blocks (device interface) to low-level Flash blocks and pages

Reduce write amplification

Ratio of the total write traffic in bytes issues by the flash chip by the FTL divided by the total write traffic issued by the OS to the device

Avoid wear out

A single block should not be erased too often

FTL – Two Systems Principles

FTL uses *indirection* and copy-on-write

Maintains mapping tables in DRAM

- Map virtual block numbers (which OS uses) to physical page numbers (which flash mem. controller uses)
- Can now freely relocate data w/o OS knowing

Copy on Write/ Log-structured FTL

- Don't overwrite a page when OS updates its data
- Instead, write new version in a free page
- Update FTL mapping to point to new location

FTL Example

Initial State

Mapping Table:			
Block 0		Block 1	
E	E	E	E

Write(a0)

Mapping Table a0->0,0			
Block 0		Block 1	
a0			
V	E	E	E

Write(a1)

Mapping Table: a0->0,0/a1->0,1			
Block 0		Block 1	
a0	a1		
V	V	E	E

Write(a1)

Mapping Table: a0->0,0/a1->1,0			
Block 0		Block 1	
a0	a1	a1	
V	V	V	E

Write(a0)

Mapping Table: a0->1,1/a1->1,0			
Block 0		Block 1	
a0	a1	a1	a0
V	V	V	V

Garbage Collect

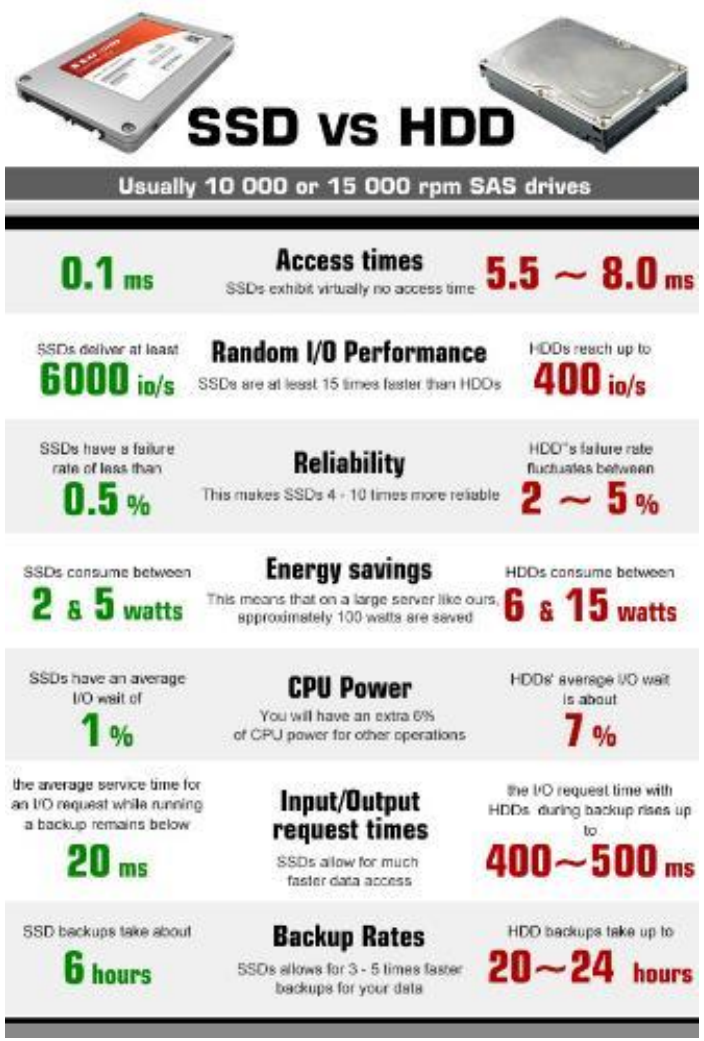
Mapping Table: a0->1,1/a1->1,0			
Block 0		Block 1	
		a1	a0
E	E	V	V

Examples of Recent SSDs

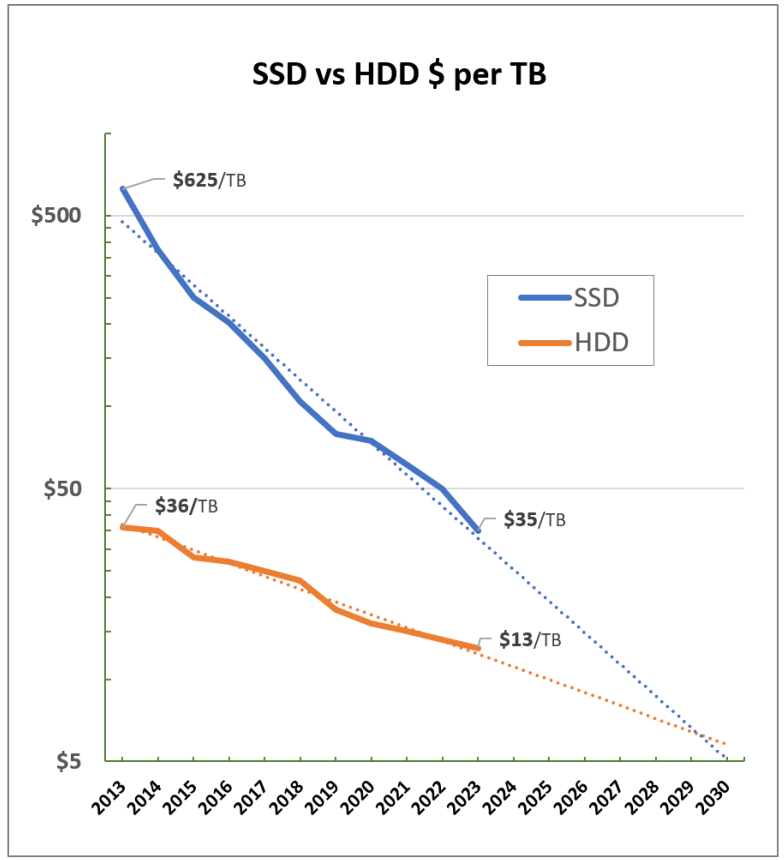
- Silicon Power 4TB **SATA** Internal SSD (2023)
  - Seq reads 540 MB/s
  - Seq writes 500 MB/s
  - Price (Amazon): $184 ($0.046/GB)

- Micron 9300 Pro 15TB **NVMe (PCIe)** Enterprise SSD (2019)
  - Seq reads/writes: 3500 MB/s
  - Random Read Ops (IOPS): 100K+
  - Price: $1750 ($0.11/GB)

HDD vs. SSD Comparison



HDD	SDD
Require seek + rotation	No seeks
Not parallel (one head)	Parallel
Brittle (moving parts)	No moving parts
Random reads take 10s milliseconds	Random reads take 10s microseconds
Slow (Mechanical)	Wears out
Cheap/large storage	Expensive/smaller storage

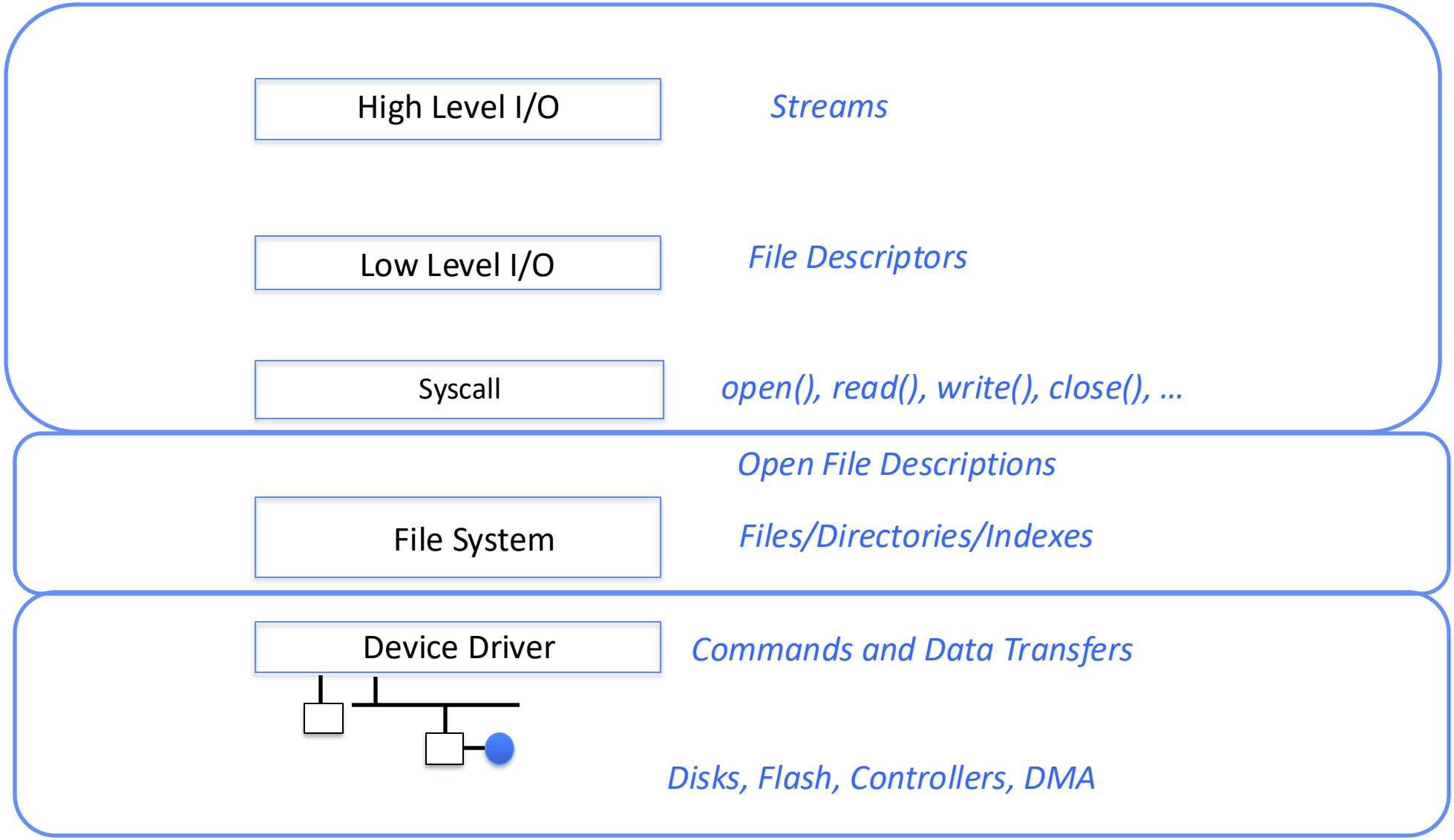


SSD prices falling faster than HDD!

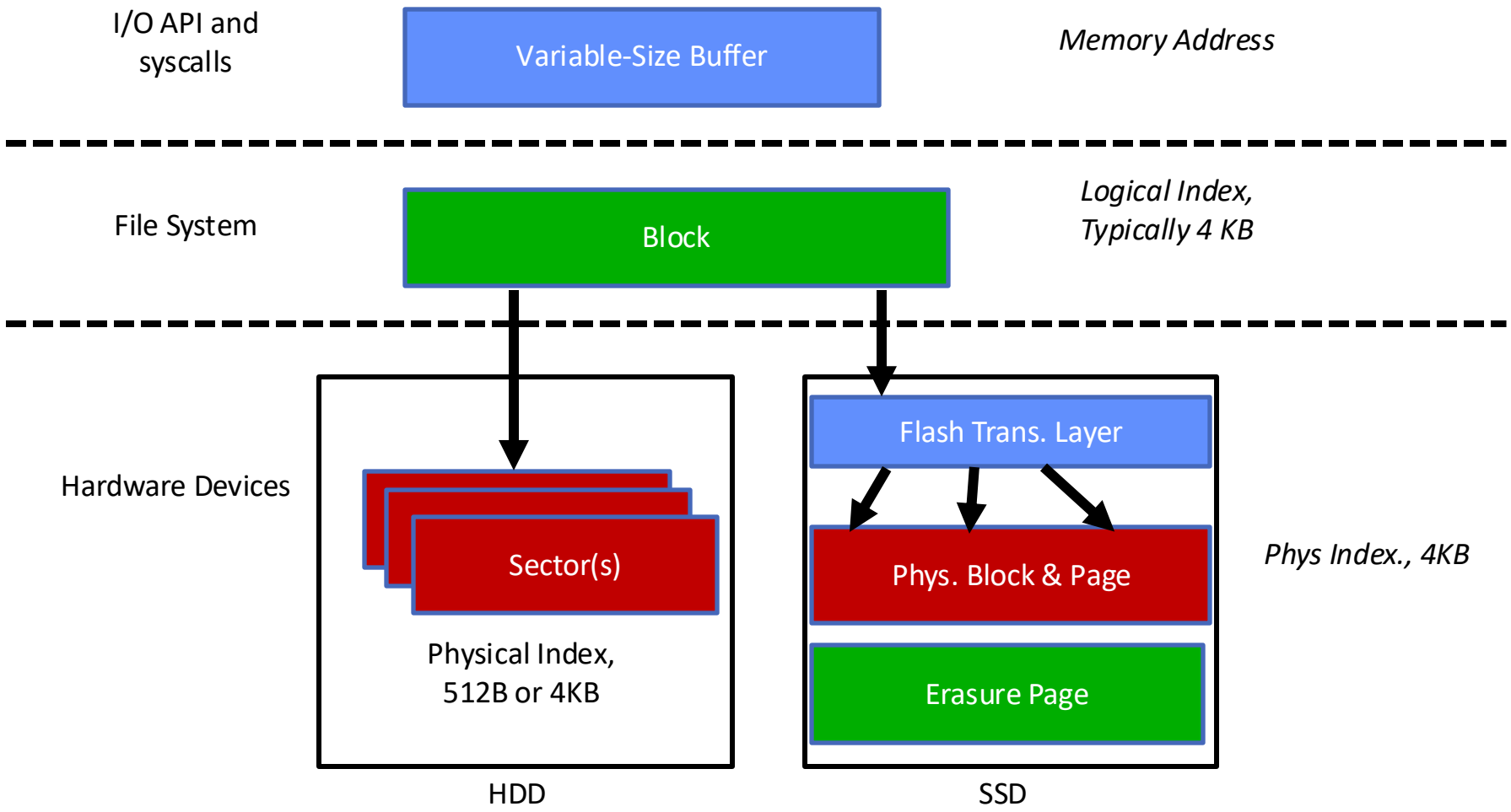
SSD Summary

- Pros (vs. hard disk drives):
  - Low latency, high throughput (eliminate seek/rotational delay)
  - No moving parts:
    - » Very light weight, low power, silent, very shock insensitive
  - Read at high speeds (limited by controller and I/O bus)
- Cons
  - Expensive (2-10x disk)
  - Asymmetric block write performance: read pg/erase/write pg
    - » Controller garbage collection (GC) algorithms have major effect on performance
  - Limited drive lifetime
    - » 1-10K writes/page for MLC NAND
    - » Avg failure rate is 6 years, life expectancy is 9–11 years
- These are changing rapidly!

Recall: I/O and Storage Layers



From Storage to File Systems



Building a File System

Layer of OS that transforms block interface of disks (or other block devices) into Files, Directories, etc.

Building a File System

OS as an illusionist:
Take limited hardware interface (array of blocks) and provide a more convenient/useful interface with:

Naming: Find file by name, not block numbers

Organize file names with directories

Organization: Map files to blocks

Protection: Enforce access restrictions

Reliability: Keep files intact despite crashes, failures, etc.

User vs. System View of a File

User's view:

- Durable Data Structures

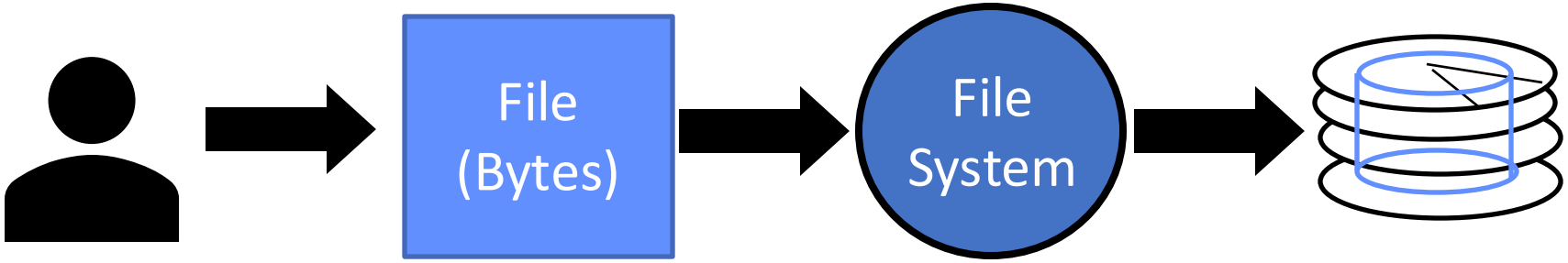
System's view (system call interface):

- Collection of Bytes (UNIX)
- Doesn't matter to system what kind of data structures you want to store on disk!

System's view (inside OS):

- Collection of blocks (a block is a logical transfer unit, while a sector is the physical transfer unit)
- Block size $\geq$ sector size; in UNIX, block size is 4KB

Translation from User to System View



What happens if user says: "give me bytes 2 – 12?"

– Fetch block corresponding to those bytes

– Return just the correct portion of the block

What about writing bytes 2 – 12?

– Fetch block, modify relevant portion, write out block

Everything inside file system is in terms of whole-size blocks

Disk Management

Basic entities on a disk:

File: user-visible group of blocks arranged sequentially in logical space

Directory: user-visible index mapping names to files

The disk is accessed as linear array of sectors

Old: Physical Position [cylinder, surface, sector]

New: Logical Block Addressing (LBA)

Every sector has integer address

Controller translates from address ⇒ physical position

Shields OS from structure of disk

What Does the File System Need?

Track free disk blocks

– Need to know where to put newly written data

Track which blocks contain data for which files

– Need to know where to read a file from

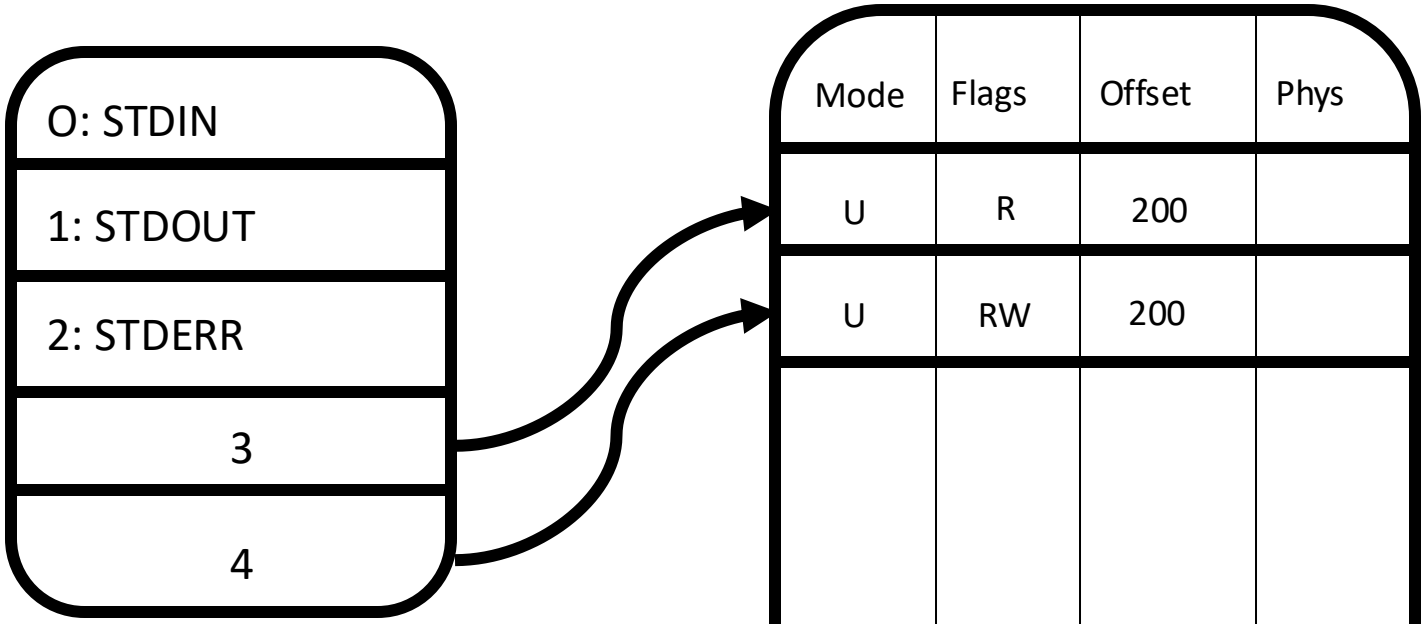
Track files in a directory

– Find list of file's blocks given its name

Where do we maintain all of this?

– Somewhere on disk

Recall: FD & File Descriptors



Per-Process File Descriptor Table

Global Open File Description Table

Critical Factors in File System Design

(Hard) Disk Performance !!!

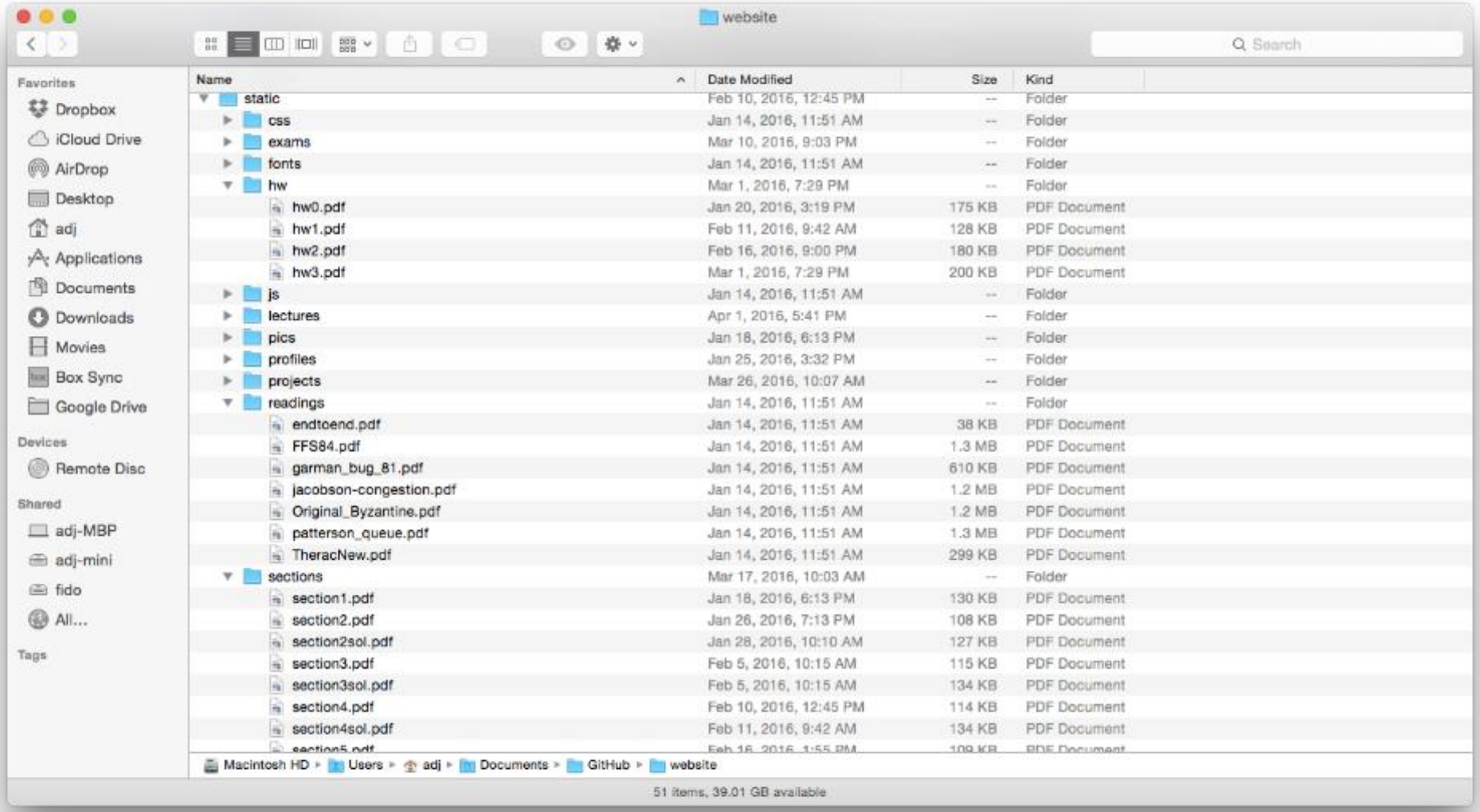
Open before Read/Write

Size is determined as they are used !!!

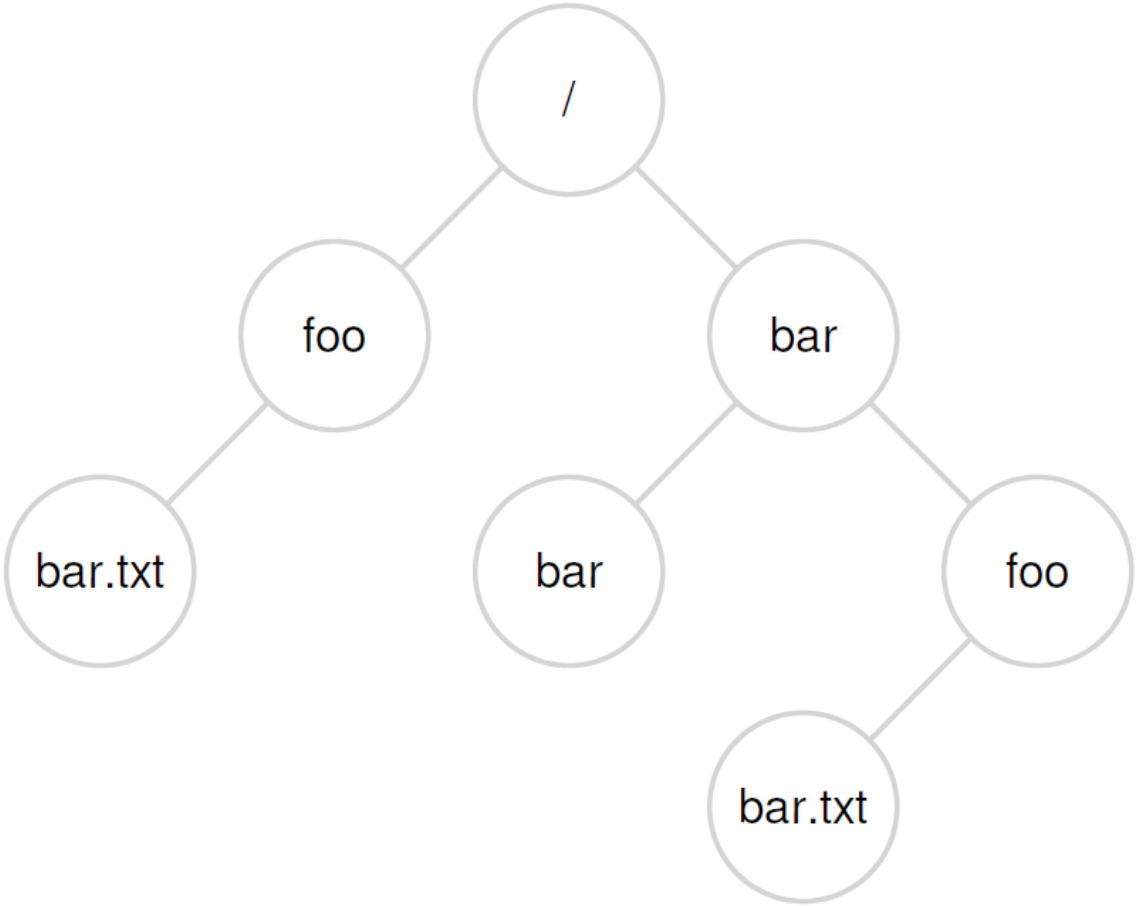
Organized into directories

Need to carefully allocate / free blocks

Files & Directories



Files & Directories



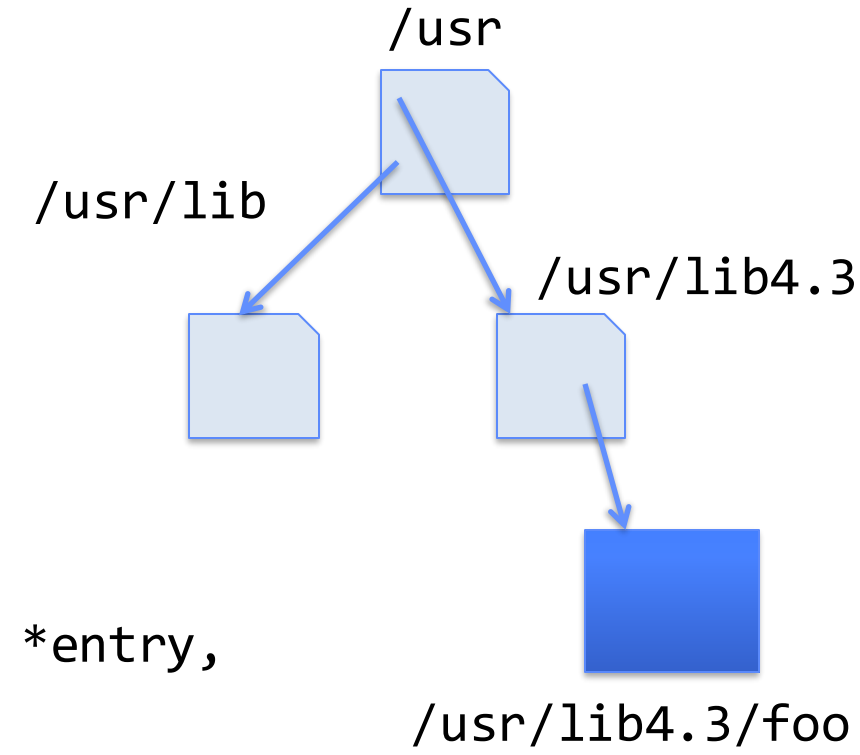
Manipulating directories

System calls to access directories

- open / creat / readdir traverse the structure
- mkdir / rmdir add/remove entries
- link / unlink (rm)

libc support

- DIR * opendir (const char *dirname)
- struct dirent * readdir (DIR *dirstream)
- int readdir_r (DIR *dirstream, struct dirent *entry,
  struct dirent **result)



Components of a File System

Superblock object: information about file system

Free bitmaps: what is allocated/not allocated

Inode object: represents a specific file

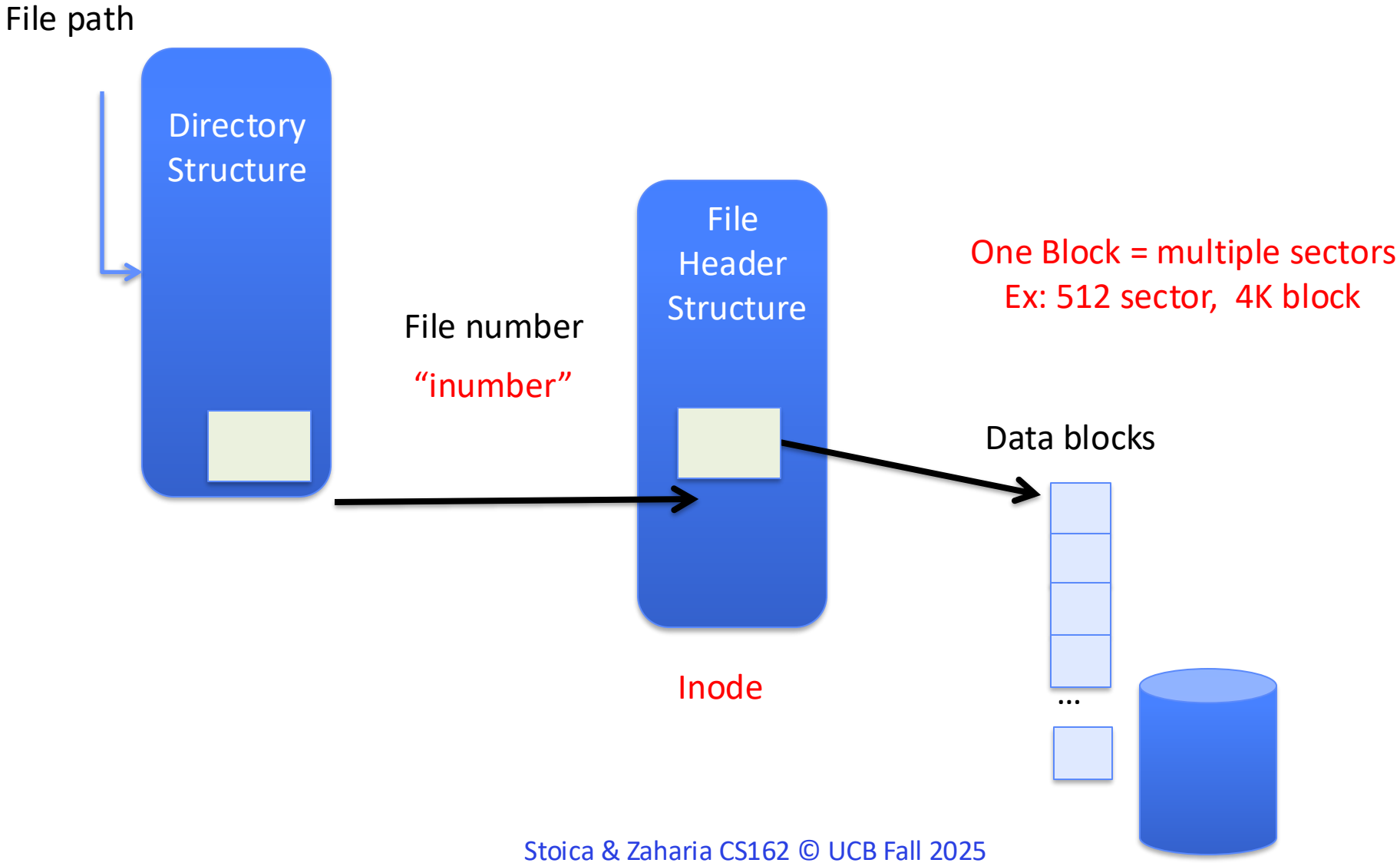
Dentry object: directory entry, single component of a path

File object: open file associated with a process.

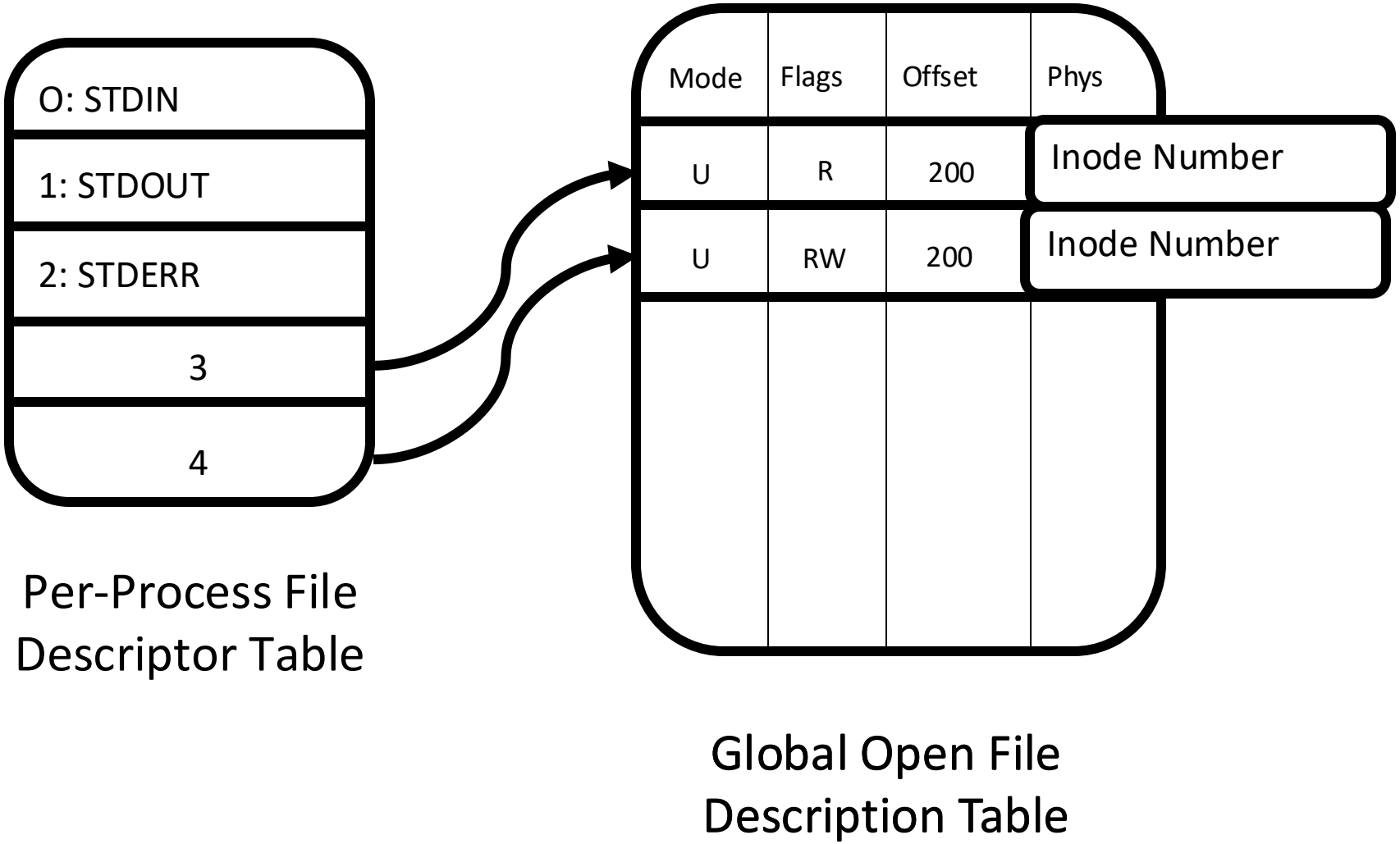
Blocks: How files are stored on disk

Components of a File System

```
open(/laptop/Natacha/cs162/foo.txt)
```



The (In)famous Inode



How to get the Inode number?

Look up in directory structure

Directory is a specialised file containing
<file_name : inode number> mappings

File number could be a file or another directory

Each <file_name : inode> mapping is called a directory entry

How to read a file from disk

Let's read file /foo/bar.txt (Time goes downwards)

	data bitmap	inode bitmap	root inode	foo inode	bar inode	root data	foo data	bar data [0]	bar data [1]	bar data [2]
open(bar)			**read**							
						read				
				read						
							read			
					read					
read()								**read**		
					write					
read()					**read**					
									read	
					write					
read()					**read**					
										read
					write					